Nijat Mahmudov

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PROFESSIONAL SUMMARY

Electrical Engineering student with background in FPGA-based digital systems, embedded systems, SNNs, and neuromorphic computing. Experienced in designing and implementing hardware accelerators and neural processing systems, with a growing focus on computational neuroscience and neural population dynamics. Interested in understanding biological neural systems and translating principles of neural computation into efficient hardware and neuromorphic architectures.

PROFESSIONAL EXPERIENCE

Imperial College London — May 2026 – Present
Summer Intern, Kozlov Lab, Department of Bioengineering — *London, United Kingdom*

- Developing a hybrid CNN–ViT for CIFAR-10 classification with a differentiable gating layer that learns adaptive ANDand OR-like feature integration
- Implemented a reproducible PyTorch training and evaluation pipeline and conducted experiments across five random seeds.
- Achieved $89.1 \pm 0.4\%$ test accuracy, outperforming CNN, ViT, and conventional CNN–ViT baselines.
- Demonstrated through ablation studies a 1.3–1.7 percentage-point improvement over fixed-operation variants with <5parameter overhead.

Armes Neuroengineering — Jun 2025 – Aug 2025
Embedded Systems Engineer Intern — *Istanbul, Türkiye*

- Profiled and optimized embedded C firmware interfacing with a neuromorphic silicon chip, targeting communication latency, memory usage, and CPU load.
- Reduced host-to-chip command latency by 38 % through buffer-management, packet-handling, and control-flow optimizations.
- Refactored the Python API and built unit and integration test suites, enabling reproducible hardware experiments and maintainable firmware integration

Max Planck Institute for Human Cognitive and Brain Sciences — Jun 2024 – Sep 2024
Research Intern, Neural Data Science Group — *Leipzig, Germany*

- Developed a modular, physics-informed MRI simulation pipeline generating images and multi-coil k-space from quantitative maps with configurable TR, TE, FA, and TI.
- Modeled realistic acquisition effects, including B0/B1 inhomogeneities, ESPIRiT coil sensitivities, correlated complex noise, and ROMEO-based phase unwrapping.
- Generated large-scale high-resolution datasets with flexible ground truth for undersampled reconstruction and segmentation tasks.
- Validated synthetic outputs against real 7T MRI acquisitions.

ITU VLSI CAD Design Laboratory — Dec 2023 – May 2024
Undergraduate Research Assistant — *Istanbul, Türkiye*

- Designed a real-time FPGA pipeline for 128×128 DVS gesture recognition, integrating AER decoding, timestamp-based time surfaces, and BRAM-based event storage
- Implemented eight parallel distance-processing units with 16-bit fixed-point inference in SystemVerilog on a Xilinx Zynq platform
- Validated the architecture against 1,342 recordings spanning 11 gestures, 29 subjects, and 3 illumination conditions, analyzing synthesis, timing, resource utilization, latency, and dynamic power at 100 MHz

Analog Devices — Jun 2023 – Aug 2023
Digital Design / FPGA Engineering Intern — *Istanbul, Türkiye*

- Redesigned timing-critical FPGA/SoC datapaths using asynchronous Sutherland-style micropipelines, AXI4-Stream interfaces, and high-speed Galois counters
- Built functional and UVM-like verification environments and executed formal protocol checks under aggressive timing constraints
- Increased the internal prototype’s maximum operating frequency from approximately 480 MHz to over 800 MHz, a 67% improvement

PROJECTS

FPGA-Based Event-Driven Biosignal Processing Platform with a Custom Spiking Neural Network Accelerator

Undergraduate Thesis

- Designed a custom LIF-SNN accelerator in SystemVerilog on Xilinx Zynq with dual-port BRAM, 64-channel spike FIFO, and runtime-configurable neuron/memory parameters via configuration registers
- Integrated threshold-based spike encoder for direct EEG/EMG input and benchmarked accuracy, latency, and dynamic power at 100 MHz

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CLANE – Spatiotemporal Action Recognition on Neuromorphic Hardware

- Developed an end-to-end on-chip spiking pipeline combining a 2D SCNN with a CLP-SNN head on Loihi 2
- Enabled online class-incremental learning from event-camera streams at approximately 5 mJ per inference
- Achieved 16$\times$ lower latency and lower energy consumption versus a Jetson Orin Nano baseline

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Neural Population Dynamics in the Mouse Visual Cortex
Independent Project

- Analyzed multi-neuronal spike-train recordings from mouse primary visual cortex using Poisson Generalized Linear Models (GLMs) in Python.
- Modeled temporal dependencies in neural population activity and evaluated model performance using held-out test data.
- Investigated how statistical dependencies in population activity can be used to characterize interactions and temporal dynamics within neural ensembles.

EEG-Based Brain-Body Interface for GVS Control
Independent Project

- Built a real-time BBI classifying EEG motor imagery (left/right) and converting output into galvanic vestibular stimulation (GVS) via wireless UDP
- Achieved above-chance classification accuracy with sub-300ms system latency
- Designed a controlled experimental protocol (sham condition, IMU-based sway measurement, statistical analysis)

EDUCATION

Istanbul Technical University — Jun 2021 – Jun 2026
B.Sc. in Electrical Engineering — *Istanbul, Türkiye*
GPA: 3.30 / 4.00
Class Rank: Top 10% of class

Türk Diyanet Vakfı Baku Turkish Lyceum — Sep 2016 – Jun 2021
High School Diploma | GPA: 5.00 / 5.00 — *Baku, Azerbaijan*

CERTIFICATIONS & AWARDS

- Azerbaijan State Scholarship Winner – awarded annually to the top 500 students nationwide
- Cadence Design Systems – Innovus Block-Level Implementation Training
- Cadence Design Systems – Innovus Clock Concurrent Optimization for Clock Tree Synthesis Training

SKILLS

Neuromorphic Computing: SNNs, Loihi 2, event cameras, Address-Event Representation, Machine Learning

Hardware Design & Verification: RTL design, Verilog/SystemVerilog, fixed-point arithmetic, timing closure, UVM-style verification, EDA & FPGA Tools, Synopsys VCS/DC/PT

Programming: Python, C, RISC-V Assembly, Tcl

LANGUAGES

English | C1 , **Turkish** | C2, **Azerbaijani** | Native